

Investigation of Spin-Decoherence in a Figure-8 Ring to Search for Cp-Violation Sources

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Abstract

In this paper, we consider the features of the figure-8 accelerator from the spin dynamics point of view. Some of the main advantages of such a structure are: the absence of spin resonance crossing during beam acceleration and the ability to manipulate polarization in the spin-transparent mode. In this paper, we numerically verify the hypothesis that the figure-8 ring is characterized by a spin coherence time of several hours; we consider the influence of second-order deviations in phase coordinates on decoherence. The modifications of the figure-8 ring to search for a signal of axion-like particles and the electric dipole moment of light nuclei are also presented.

Introduction

One of the aims for a polarized beam machine is to preserve beam polarization during acceleration. In a conventional magnetic ring integer and betatron spin resonances are present, crossing of the latter leads to polarization loss. One way to avoid resonance crossing is to operate at a fixed spin-tune or in the vicinity of a spin resonance. Such mode of storage ring operation is called spin-transparent (ST). In this case polarization direction is repeated after one turn. One of the benefits of the ST regime is that

polarization can be stabilized in the desired direction by elements with small field integrals – spin navigators [1].

One way to realize the ST concept is to install siberian snakes in the machine. To cover the entire energy range of a polarized beam booster one should use solenoidal snakes at low energies and snakes with transverse fields at high energies. Another issue with snakes is that for particles with low value of magnetic anomaly G , e.g., deuterons, that field integrals in snakes become challenging.

Another way to make a ST configuration is to use the natural topology of ST ring – figure-8 configuration. For the latter the total spin rotation angle is zero for arbitrary particle energy. That indicates that one can expect long spin coherence time in such structure [2].

Spin-decoherence in a figure-8 ring

Spin decoherence is the consequence of spin precession frequency spread of particles in the beam. For high precision experiments with polarized beams spin decoherence should be minimized to investigate spin dynamics in the ring plane.

In a purely magnetic ring the average energy level shifts for particles with deviations in 6D phase space. This effect causes spin-decoherence [3]. This is a second-order effect in phase space deviations and it can limit the theoretically possible spin coherence time (SCT) of several hours [2] predicted in the first-order approach for the figure-8 ring.

If we examine spin rotation in a magnetic ring then spin-tune deviation is: $\Delta\nu_s = G\Delta\gamma$. Here G is particle magnetic anomaly and γ is Lorentz factor. The linear part in $\Delta\gamma$ can be canceled with the use of an RF cavity. Then comes the energy shift caused by second-order phase deviations [3]:

$$\Delta\gamma = \frac{\beta_s^2 \gamma_s^3}{\gamma_s^2 \alpha_0 - 1} \left[\frac{\delta_m^2}{2} \left(\alpha_1 + \frac{3}{2} \frac{\beta_s^2}{\gamma_s^2} - \frac{\alpha_0}{\gamma_s^2} + \frac{1}{\gamma_s^4} \right) + \left(\frac{\Delta L}{L} \right)_\beta \right]. \quad (1)$$

Here $\delta_m = \frac{\Delta p_m}{p_s}$ – is the amplitude of synchrotron oscillations for a particle with momentum offset $\Delta p_m = p_m - p_s$ from the reference value p_s and $\left(\frac{\Delta L}{L}\right)_\beta$ is betatron orbit lengthening. α_0 and α_1 – first and second order momentum compaction factor.

The hypothesis to be verified is that the effect of synchrotron and betatron motion on decoherence is negligible in a figure-8 ring.

Basing on the formalism above, spin dynamics of a particle with 6D phase deviations can be regarded as motion with an average energy shift $\Delta\gamma$. The total spin rotation angle Ψ is:

$$\Psi = \Phi_P \cdot (\gamma + \Delta\gamma)G + (-\Phi_P) \cdot (\gamma + \Delta\gamma)G = 0, \quad (2)$$

where Φ_P is the momentum rotation angle in the arcs.

In this formulation $\Psi = 0$ in a figure-8 ring for arbitrary energy shift. The numerical experiment indicates the same result.

For the numerical experiment an assumption was made that the lattice is ideal, i. e. has no alignment errors of the elements. This is due to the fact that the coherent part of the resonance strength caused by the misalignments can be compensated in the synchrotron with high precision implementing spin response function formalism [4, 5]. The incoherent part of the resonance strength, related to phase space deviations, remains and its influence on decoherence is investigated.

The investigation is carried out for the deuteron beam at 240 MeV kinetic energy, that is optimal for the polarimeter analyzing power [6]. The typical beam parameters are $\varepsilon_{x,y} = 1 \text{ mm}\cdot\text{mrad}$ and $\delta = 1\text{e-}4$. The numerical experiment is performed with COSY Infinity software [7]. Resonant structure optics of a figure-8 ring was used in the simulations [8]. The spin transfer matrix is unity for the reference particle in the figure-8, RF is ON, natural chromaticity of the ring is not corrected with sextupoles.

The first thing to verify was the correctness of orbit lengthening effect calculation in a figure-8 with COSY Infinity. Coordinate and energy-offset particles were considered (Fig. 1, 2).

For particles with momentum and transverse phase space offsets the analytical formulae for the orbit lengthening are [9]:

$$\frac{\Delta L}{L} = \alpha_0 \cdot \delta + \alpha_1 \cdot \delta^2, \quad (3)$$

$$\left(\frac{\Delta L}{L}\right)_\beta = -\frac{\pi}{L} [\varepsilon_x \xi_x + \varepsilon_y \xi_y]. \quad (4)$$

Here $\xi_{x,y}$ are beam chromaticities and $\varepsilon_{x,y}$ are the Courant-Snyder invariants.

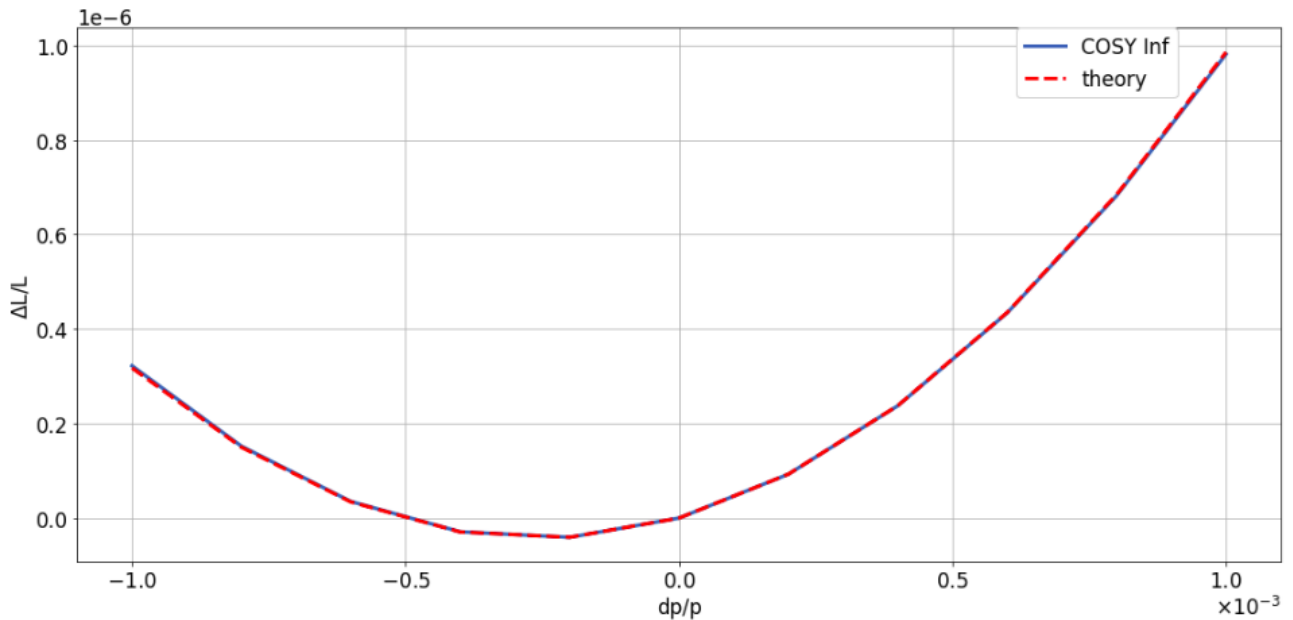


Figure 1. Orbit lengthening for off-momentum particles in a figure-8 ring. Red line – analytical result, blue line – COSY Infinity calculations.

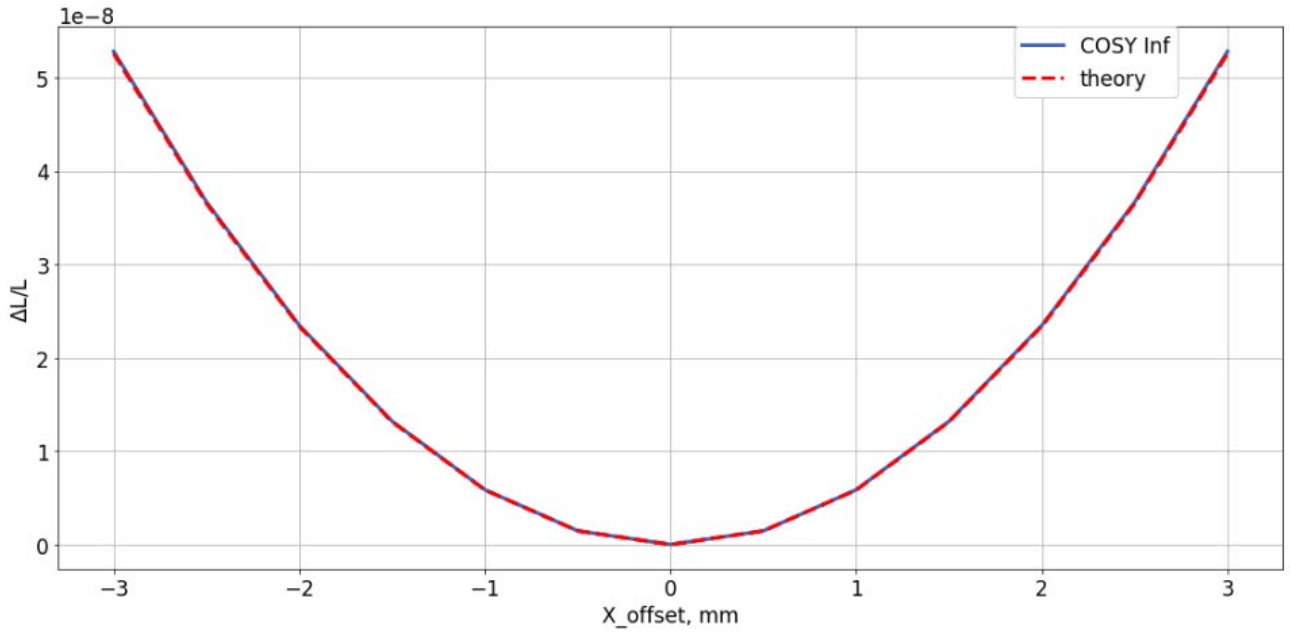


Figure 2. Orbit lengthening for particles with radial offset in a figure-8 ring. Red line – analytical result, blue line – COSY Infinity calculations.

Then spin-tracking was performed for these particles with initial spin being longitudinal ($S_z = 1$). Figures 3-5 represent projections of spin-vector with the most changes for each ray (with X, Y and δ offset). The most significant decoherence effects were observed for energy-offset particles (Figures 5). Spin-tracking was performed for 10^9 turns that corresponds to 1000 seconds of beam orbiting in the ring. This corresponds to the experimentally achieved SCT for a deuteron beam at a magnetic racetrack ring COSY [10]. The second-order decoherence effects were adjusted with sextupoles.

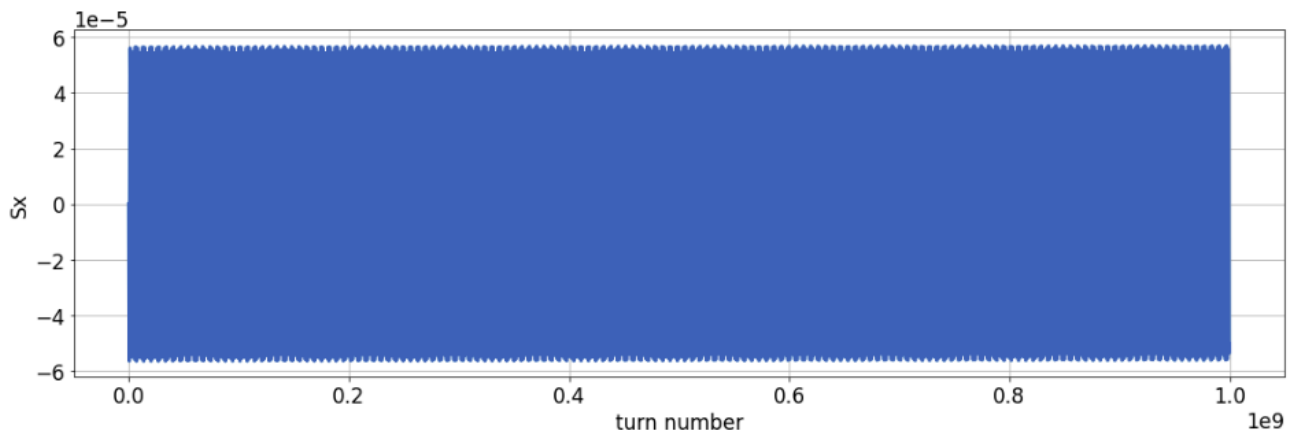


Figure 3. Spin tracking results for horizontal spin projection depending on turn number. The particle has 1 mm horizontal deviation from reference.

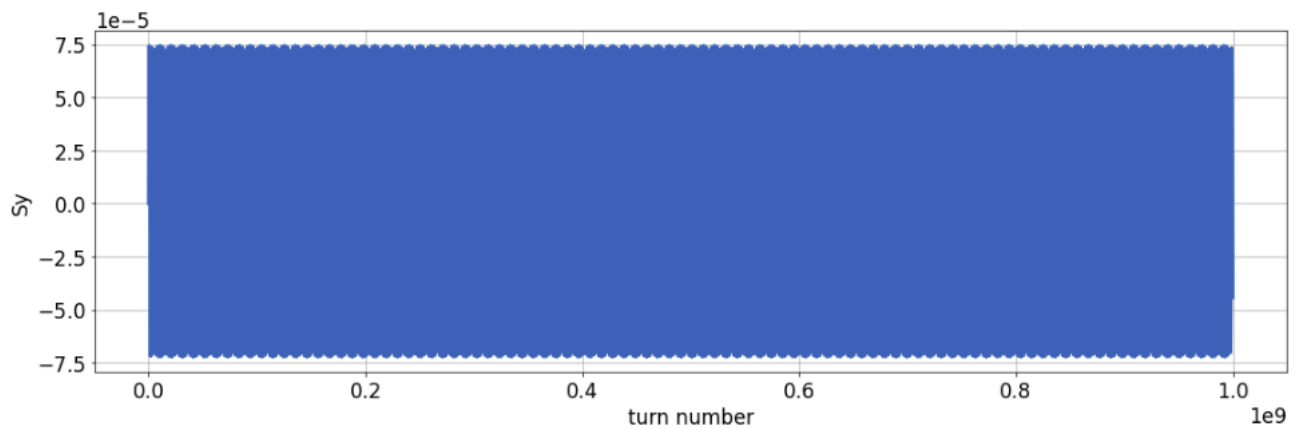


Figure 4. Spin tracking results for vertical spin projection depending on turn number. The particle has 1 mm vertical deviation from reference.

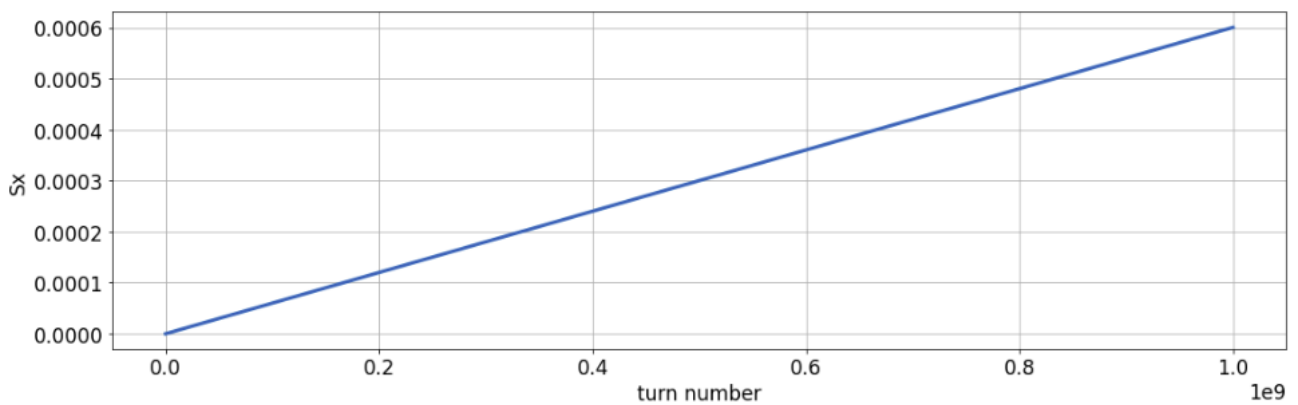


Figure 5. Spin tracking results for horizontal spin projection depending on turn number. The particle has $dp/p=10^{-4}$ deviation from reference.

Figure-8 ring for axion search

Long spin coherence time is an extremely positive condition for collecting statistical data when conducting precision experiments with polarized beams. For example, this can be an experiment to search for axion-like particles. For this purpose, it is necessary to add a Wien filter to the accelerator structure to set the spin precession frequency and to search for resonance with the axion field [11].

For low frequency of axion field WF in a ST ring acts as a perturbation. So, placing a single WF in the straight section of a figure-8 ring doesn't violate long SCT condition (Figure 6).

Figure-8 ring can be used for axion search in the low mass domain:

$$\hbar\omega_a = m_a c^2. \quad (5)$$

The benefits of low frequency axion search are presented in [2]. In this case the axion phase validity time

$$\tau_a = \frac{h}{m_a v^2} \quad (6)$$

is large as well as the SCT and possible data taking period.

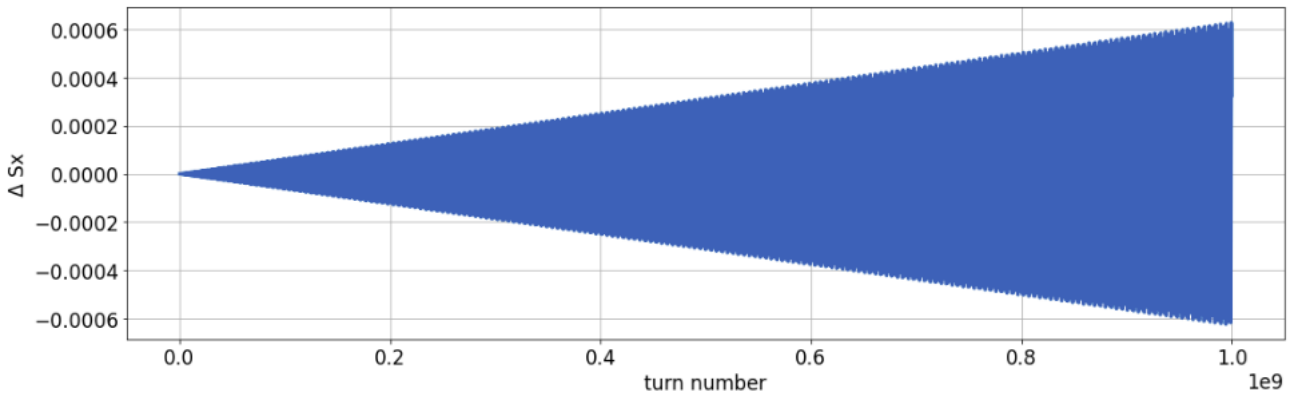
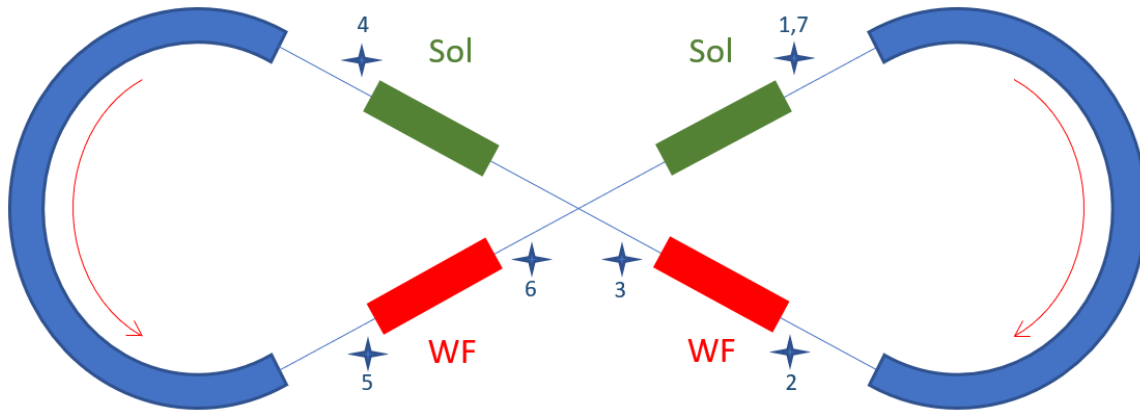


Figure 6. Spin tracking results for horizontal spin projection minus value of reference particle depending on turn number. Figure-8 ring with a WF induced spin-tune of 1e-6 is used. The particle has dp/p=1e-4 deviation from reference.

Figure-8 ring for the EDM search

In a pure figure-8 ring the measured EDM-signal is suppressed being co-directed with the Lorentz force in each of the arcs. To modify the lattice for the EDM experiment one can add Wien filters in straight sections to compensate for the spin rotation angle in the arcs to fulfil the Quasi-Frozen Spin condition [12]. Snake solenoids should be also added to allow for the accumulation of spin rotation from the EDM in the structure (Figure 7).



Horizontal plane	Vertical plane
1	1
2	2
3	3
4	4
5	5
6	6
7	7

Figure 7. Schematic view of a figure-8 ring adopted for the EDM experiment with corresponding table of spin rotations along the structure.

Spin decoherence is a key limiting factor in the EDM experiment. In this regard the EDM option of figure-8 was investigated (Figure 8).

One can state that 2 full siberian snakes in straight sections turn the figure-8 ring into a racetrack option in terms of decoherence. The spin deviation for off-momentum particles is added up after each arc of the ring in contrast to the pure figure-8 structure.

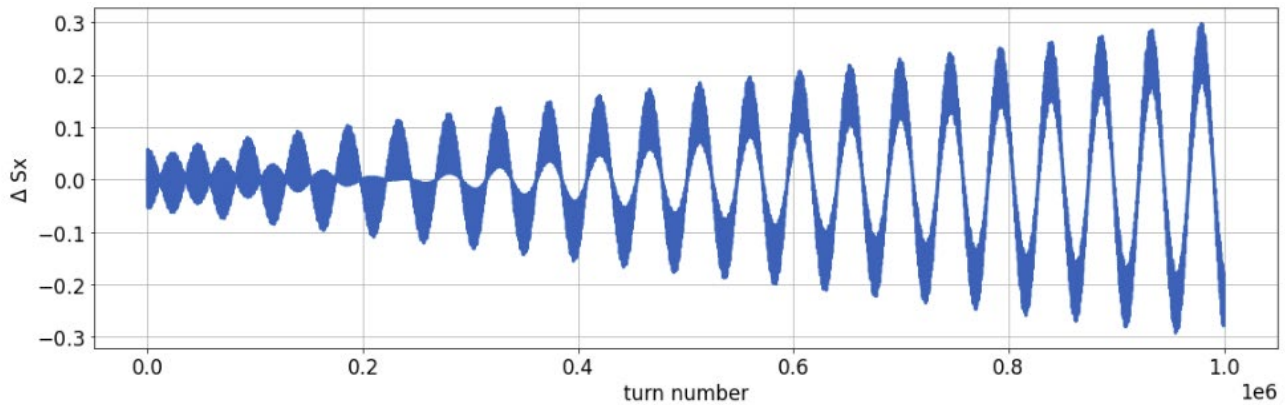


Figure 8. Spin tracking results for horizontal spin projection minus value of reference particle depending on turn number. The EDM option of figure-8 ring is used. The particle has $dp/p=1e-4$ deviation from reference.

Conclusion

The figure-8 accelerator is a multipurpose facility for polarized beam experiments. At low energies the hypothesis of long spin coherence time in a pure figure-8 holds true. The influence of synchrotron and betatron motion on decoherence can be formulated in terms of the average energy shift. The figure-8 ring can be modified to search for the axion-like particles by installing a Wien filter in the straight section. The figure-8 ring can also be adopted for the EDM search by installing 2 solenoids and Wien filters in straight sections, nevertheless, long spin coherence time is not foreseen in this scheme.

Acknowledgements

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